

Highlights from a research sprint:

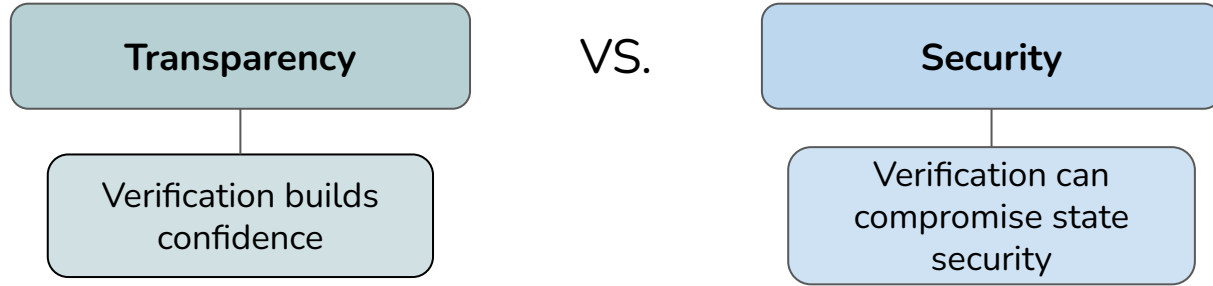
Does Stringent Verification Deter Participation?

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Verification mechanisms



How does verification stringency affect participation in arms control agreements?

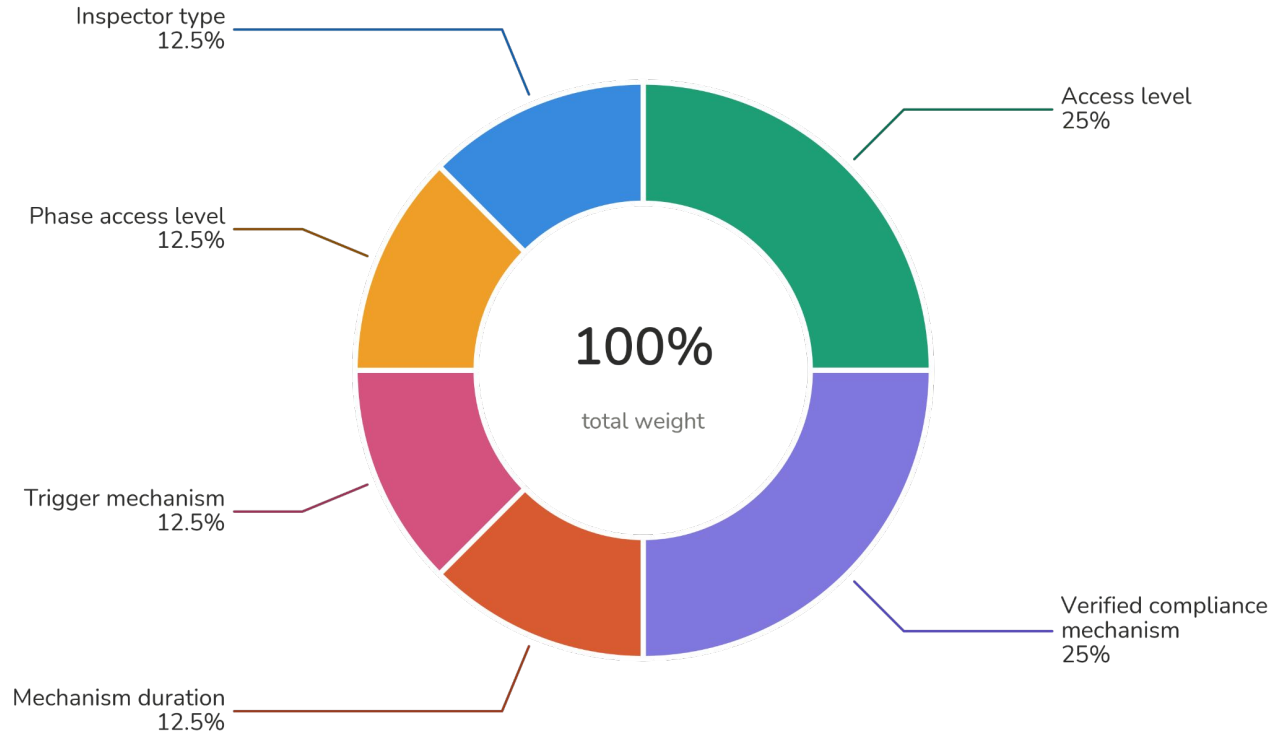
Verification Index

To assess the impact of verification stringency on ratification, we created an **index** out of six measures derived from existing variables within the amc_vercom_v2 dataset, and assigned them a 'weight' value.

Measure	Variable	Assigned Weight (=1)
Mechanism Duration	<i>verified_compliance_mechanism_duration.</i>	0.125
Trigger Mechanism	<i>verified_compliance_mechanism_trigger_type,</i> <i>verified_compliance_mechanism_agreement_trigger.</i>	0.125
Inspector Type	<i>verified_compliance_mechanism_inspector_type.</i>	0.125
Phase-Access Level	<i>verified_compliance_mechanism_[phase].</i>	0.125
Verification Mechanism	<i>verified_compliance_mechanism_category.</i>	0.25
Access Level	<i>Access_level variables weighted (1 - 2) and combined.</i>	0.25

Assigned weights by measure:

Six measures - weights sum to 1.0



Verification, Demonstration, and Consultation

- We found **31 agreements** with **at least 1** verification mechanism
- To construct a measure of verification stringency for the rest of the agreements, we included a rough assessment of the number of demonstration and consultation mechanisms, resulting in an *extended_index*

Extended Verification Stringency Index:

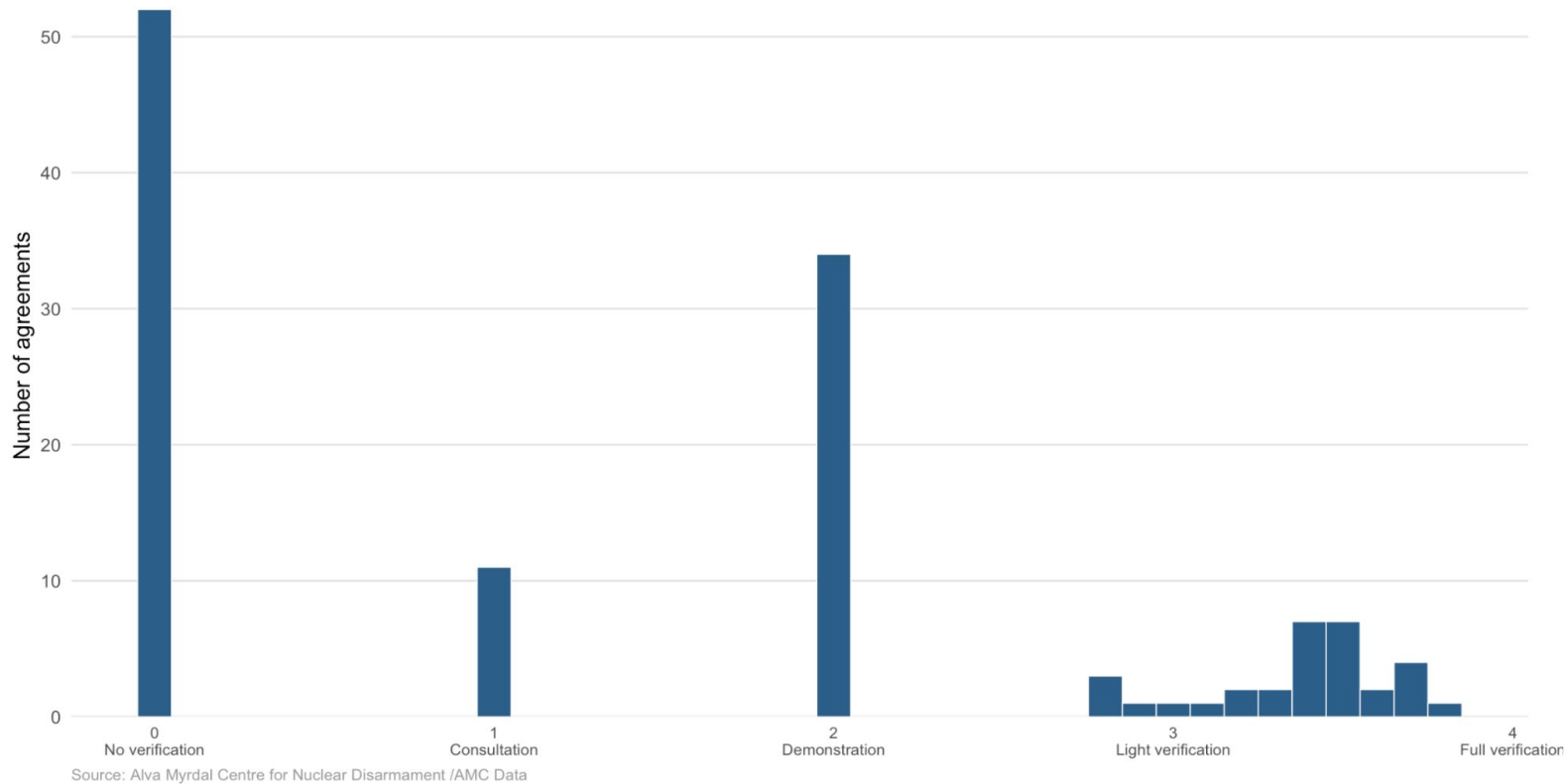
1 - *Consultation mechanism*

2 - *Demonstrated compliance mechanism*

>2 - 4 - *Index: Continuum of verification compliance (low to high)*

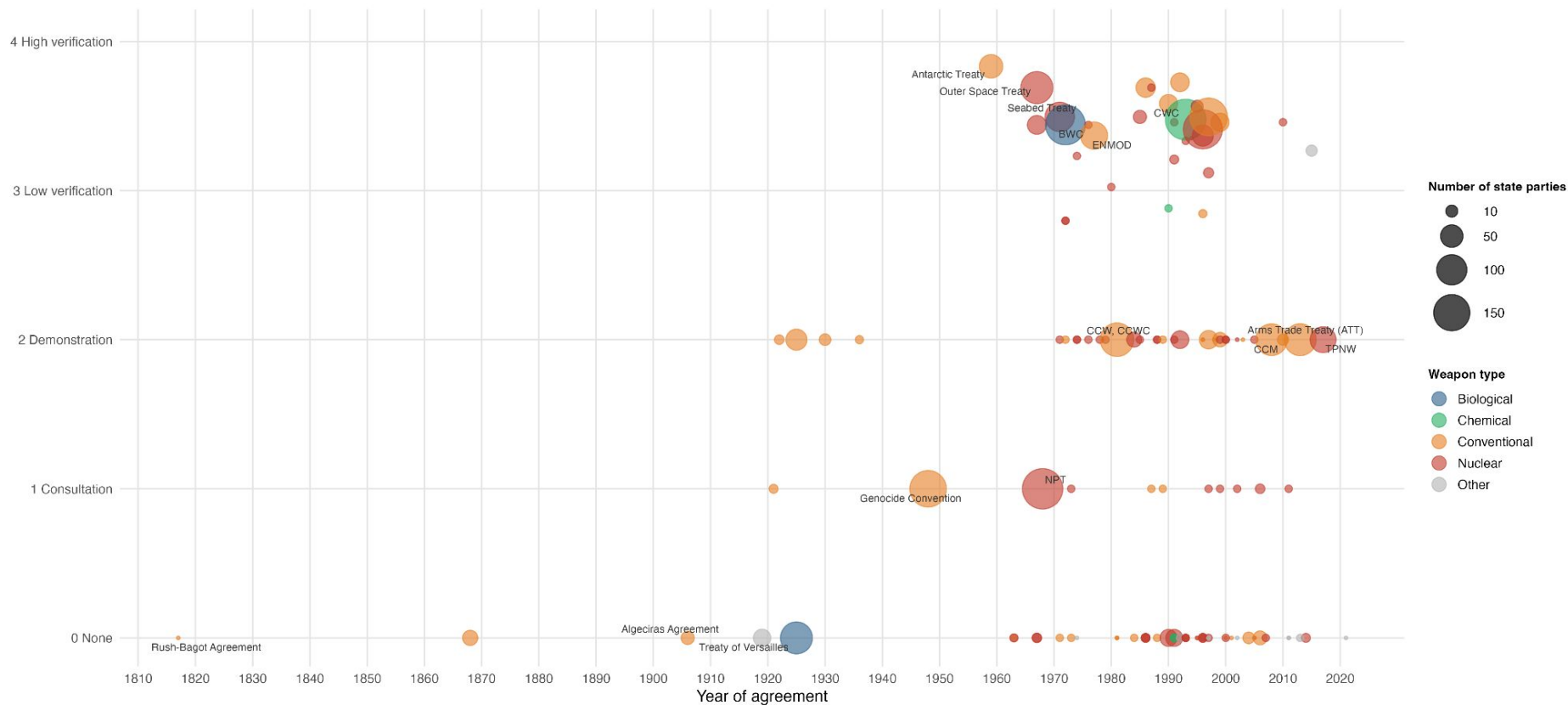
Distribution of Verification Stringency by Agreement

Based on extended verification intensity index (0-4)



Verification Stringency by Agreement over Time

Each point represents one agreement, colored by weapon type



Verification stringency and treaty participation

Then, to assess the association between verification and participation, we measure treaty participation for each agreement in a few different ways:

By number of states:

- *Number of states parties*
- *Number of participating states*

By ratification share:

- *% of participating states*
- *% of initial signatories*

Regression setup

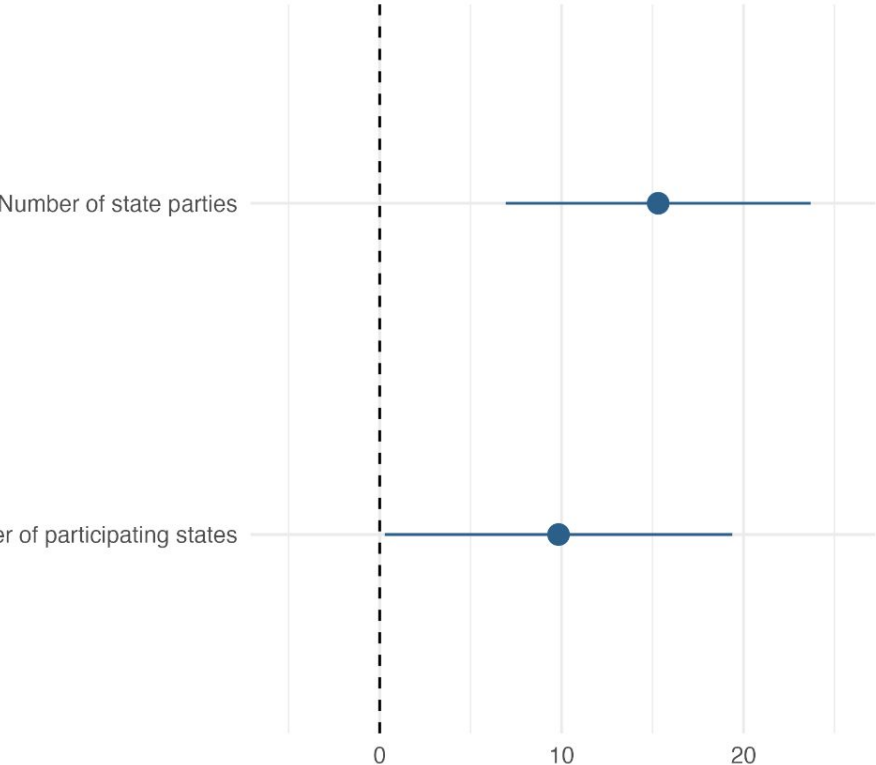
Using our constructed verification index and measures of participation, we run a linear regression model, controlling for additional confounders:

- **Weapon type** (*nuclear, chemical, biological, conventional*)
- **Scope** (*how much is covered?*)
- **Era** (*dummy for pre/post Cold War, pre/post 2001*)

Effect of Verification Stringency on Treaty Participation

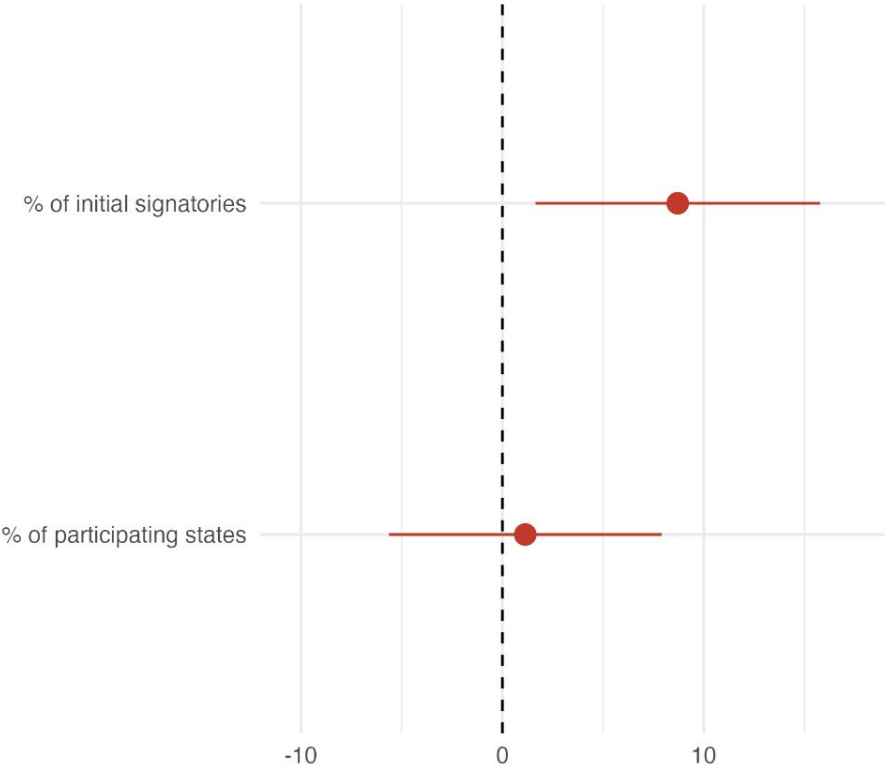
Controlling for weapon type, scope, and era | Multilateral agreements only

By number of states:



Coefficient estimate (OLS)

By ratification share:



Coefficient estimate (OLS)

Future Research - The Eligible States Denominator

- To accurately measure participation in treaties, it would be useful to measure the **states which *have not* signed agreements for which they are eligible**.
- To do so would require coding each agreement's eligible universe from its scope.
- This would enable a ***measurement of those states reluctant or which refuse to sign and ratify agreements***, to test the impact of verification measures.

From: count data; the number of states that signed, and ratified.

To: of the states eligible to join, how many did?

Insights for AI governance

Verification does not appear to deter participation in arms control agreements.



IAEA for AI ?

References

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